ModelIt!

Student Activity Pack

Can You Bend Water?

Exploring the Properties of Matter

Grade 2 | NGSS 2-PS1-1 | Patterns

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What happens to the shape of water when you pour it from a cup into a bowl?

2 Can you bend a rock? Can you bend water? What is different about them?

3 Draw what you think water looks like inside a tall cup and inside a flat plate.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Container Shape | Water Shape | Water Level

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Tall Skinny Cup

Container Shape set to tall and narrow

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Wide Flat Bowl

Container Shape set to wide and flat

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Round Jar

Container Shape set to round

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Temperature — How hot or cold the water is — cold water moves more slowly than warm water

Amount of Water — How much water you have — more water fills more of the container

Container Material — What the container is made of — some materials absorb water and some do not

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Can water bend, stretch, or change shape — and what does that tell us about matter? Use evidence from your simulation to support your answers.

1 What surprised you most about how water changes shape?

2 How is water different from a block of wood when you put them in a new container?

8 STEM Challenge: Design a Water Maze

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YOUR ENGINEERING MISSION

Build a water maze that moves water through at least three different container shapes to show how liquid changes shape.

THE SCENARIO

A water park needs a new attraction that shows visitors how cool water is! Your team has been asked to build a mini water maze that moves water through different shapes. Use what you learned about liquid properties to explain your design.

Engineering Design Thinking

1. WHAT MATERIALS WILL YOU USE TO BUILD YOUR CONTAINERS?
2. HOW WILL YOU MOVE THE WATER FROM ONE SHAPE TO THE NEXT?
3. HOW WILL YOU KEEP THE WATER FROM SPILLING BETWEEN CONTAINERS?
4. HOW WILL YOU SHOW VISITORS THAT THE WATER CHANGED SHAPE BUT THE AMOUNT STAYED THE SAME?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Materials Scientists study the properties of different materials to help make new products, from water bottles to building materials. They earn $55,000–$100,000/year.